



U.S. General Services
Administration



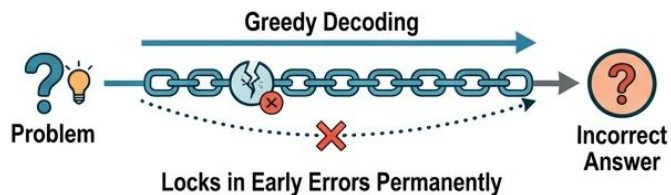
Chapter 5.3: Self-Consistency Fundamentals

US General Service Administration
AI Community of Practice - March 2026

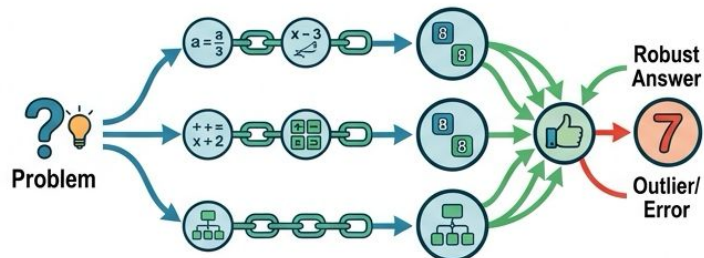
The Core Mechanism



Chain-of-Thought Commits to Its First Plausible Path



Self-Consistency: Sampling Multiple Independent Paths

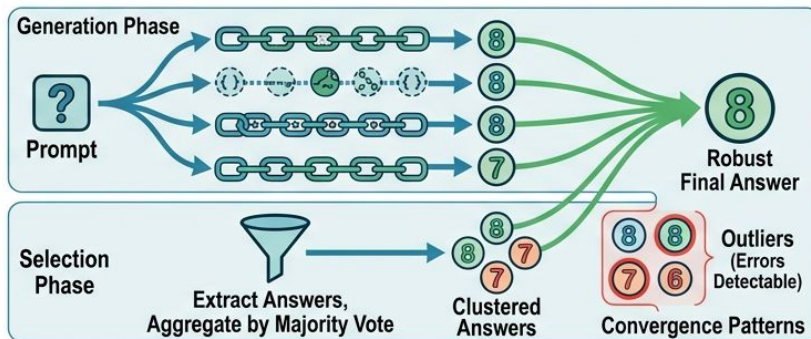


Majority Voting Selects the Most Robust Answer

The Core Mechanism:
Decoupling Generation from Selection

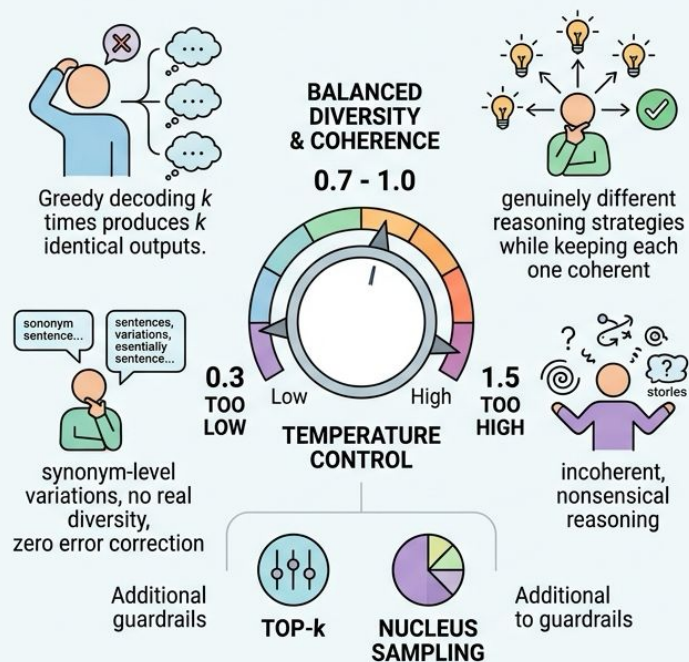


Self-Consistency: Two Distinct Phases

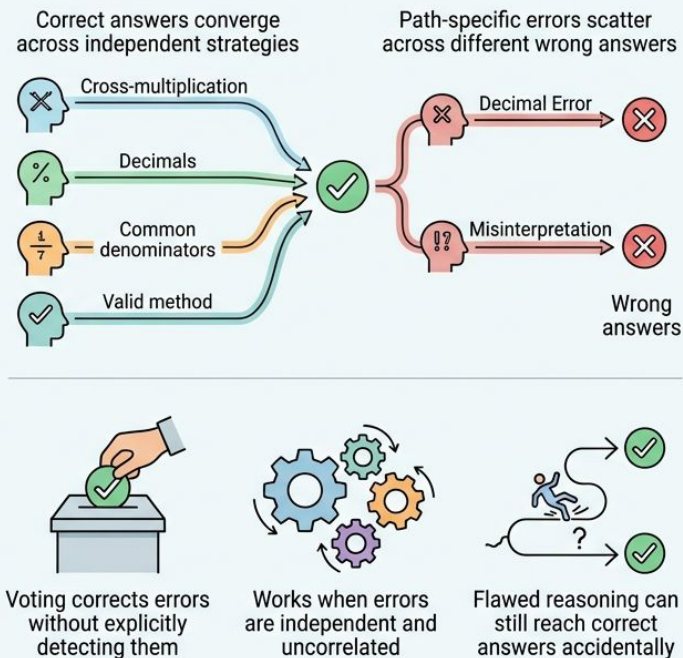


Stochastic Sampling & Majority Voting

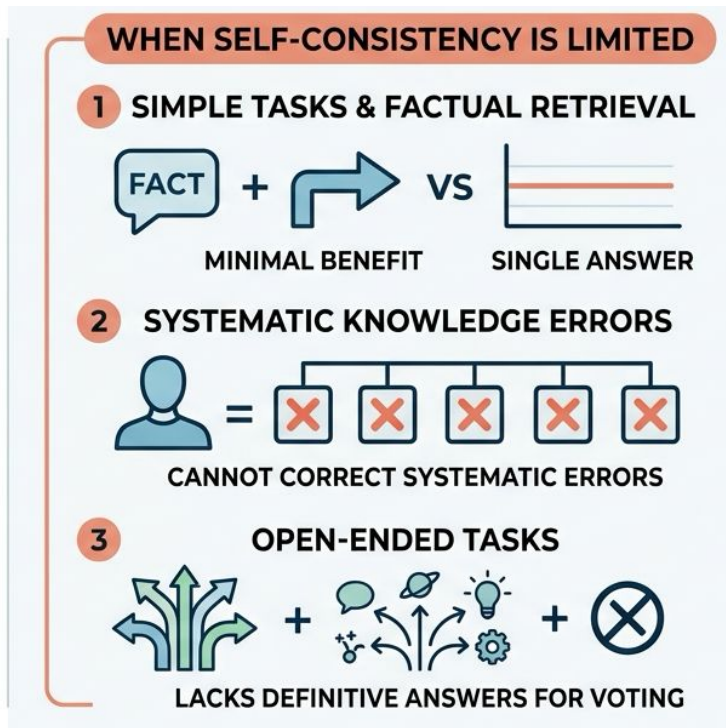
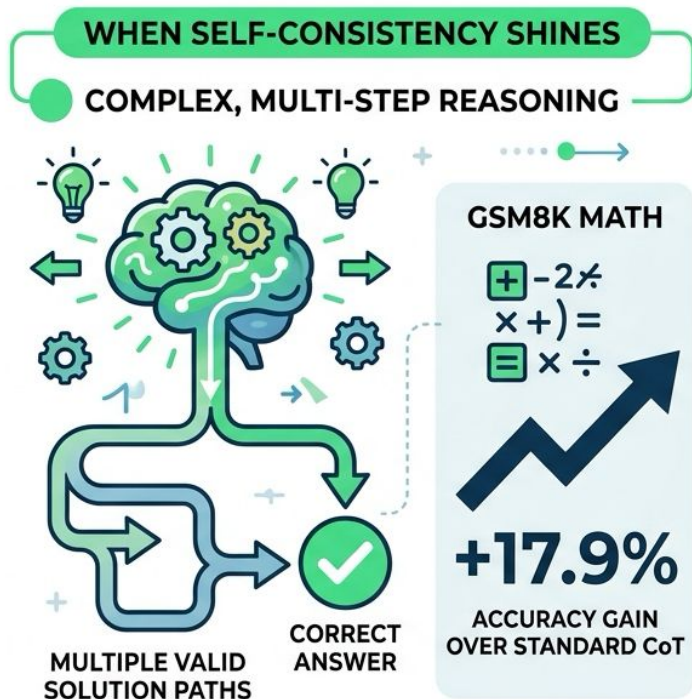
STOCHASTIC SAMPLING: CREATING MEANINGFUL DIVERSITY



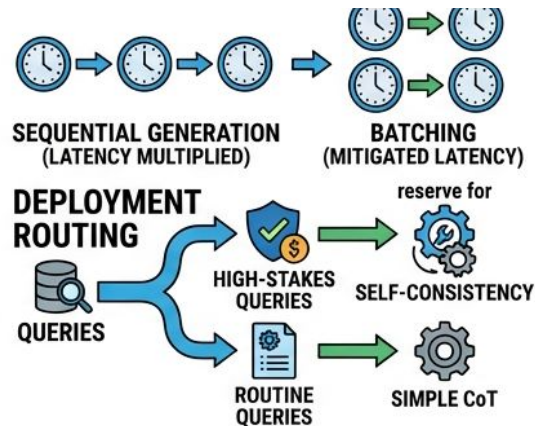
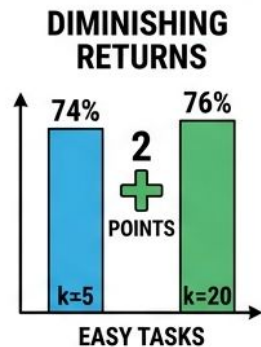
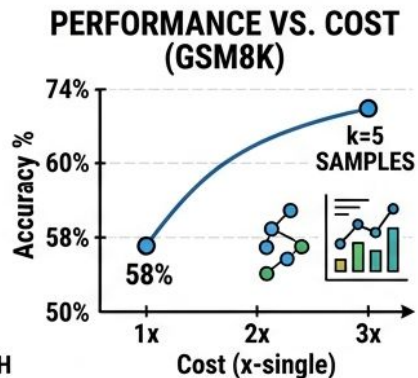
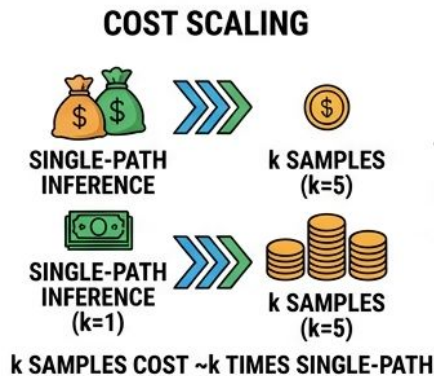
MAJORITY VOTING AS ERROR CORRECTION



When Self-Consistency Shines -- and Not Shine

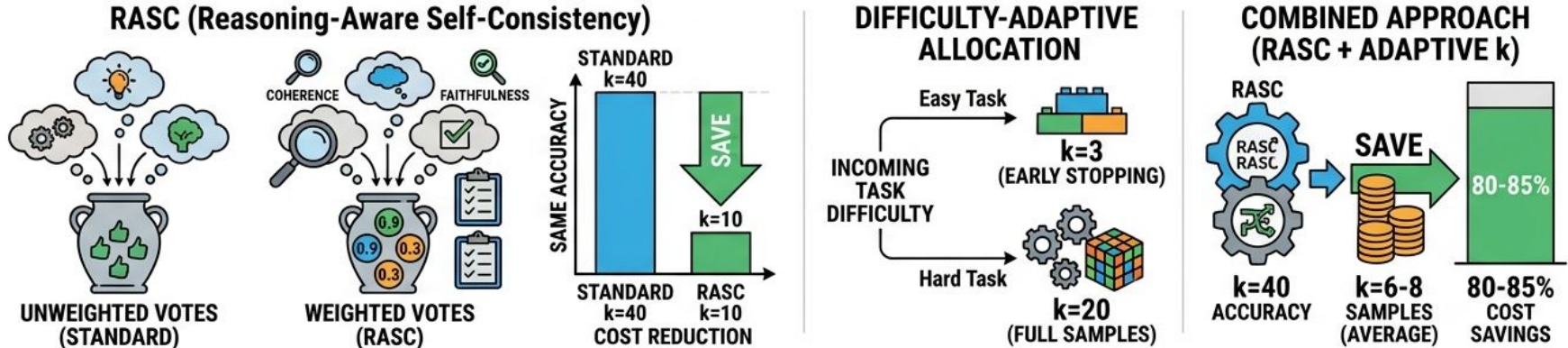


Cost-Accuracy Trade-offs in Production



- k samples cost approximately k times single-path inference
- k=5 on GSM8K: 58% to 74% accuracy at 3x cost
- Diminishing returns: k=5 to k=20 adds 2 points on easy tasks
- Latency multiplies for sequential generation (mitigated by batching)
- Reserve Self-Consistency for high-stakes queries

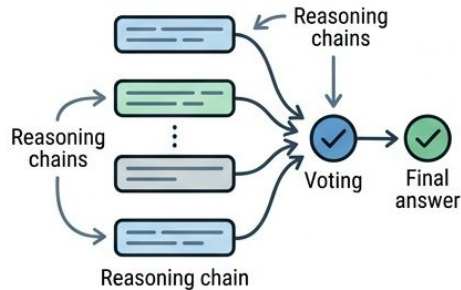
Smarter Sampling -- RASC and Adaptive Allocation



- Not all reasoning paths deserve equal votes
- RASC weights votes by coherence, faithfulness, relevance
- Achieves k=40 accuracy with only k=10 samples (70% cost reduction)
- Difficulty-adaptive allocation: k=3 for easy, k=20 for hard
- Combined approach averages k=6-8 for 80-85% cost savings

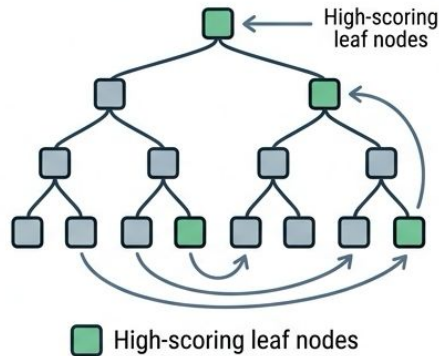
Integration with CoT, ToT, and GoT

CoT + Self-Consistency

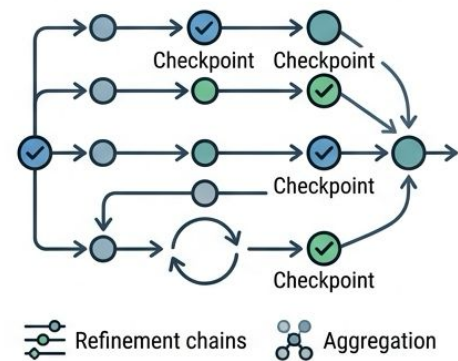


Standard layered implementation for majority of production use cases

ToT + Self-Consistency



GoT + Self-Consistency



Self-Consistency

Provides error correction



Error-Correction,
robust answers

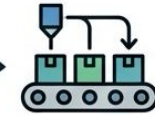


Exploration



Refinement

CoT + Self-Consistency is sufficient
for most production needs.

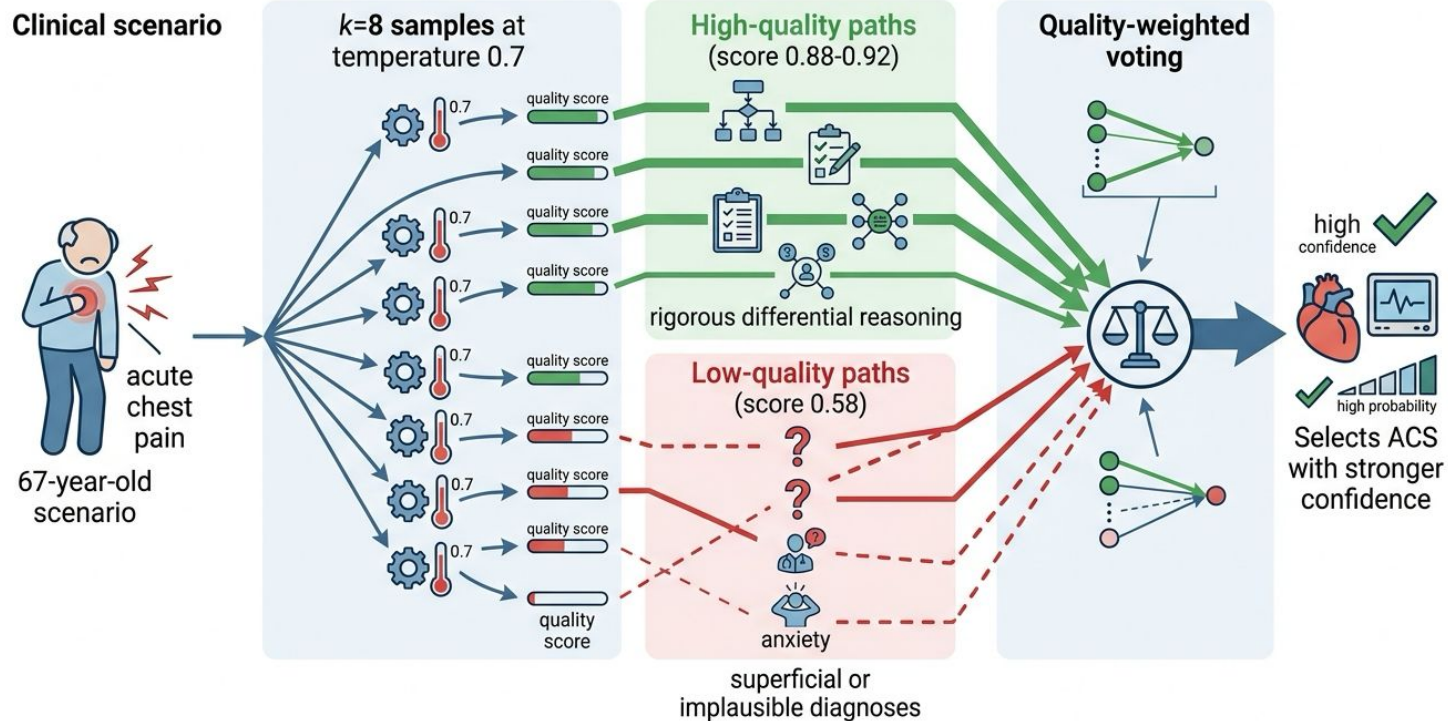


Production
assembly line

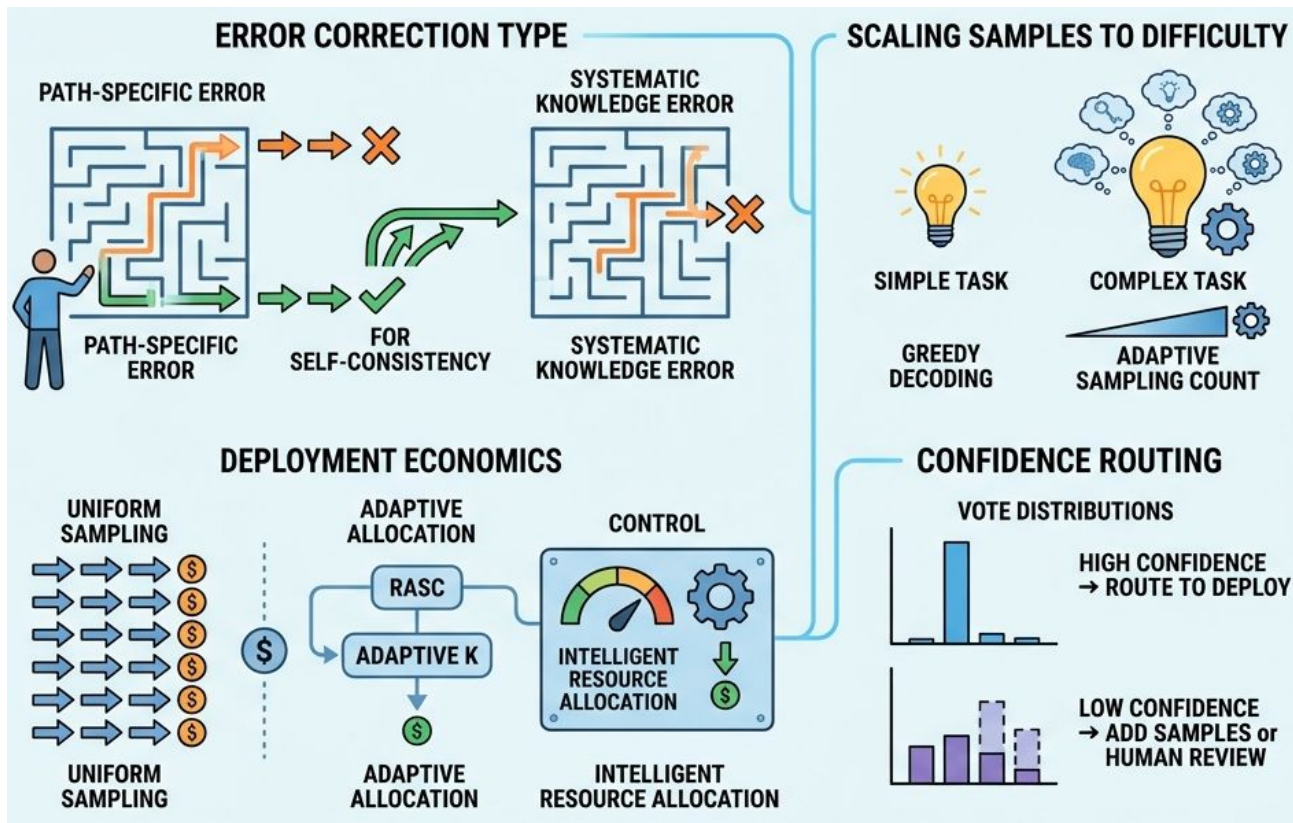


Optimal
Solution

Case: Medical Diagnosis with Weighted Voting



Optimizing Self-consistency



Key Takeaways

- Self-Consistency corrects path-specific errors, not systematic ones
 - Match sample count to problem difficulty and error economics
 - RASC and adaptive k make production deployment economical
 - Vote distributions enable intelligent confidence-based routing
 - Next: Chapter 5.4 tackles hierarchical planning for multi-step tasks
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